STOCKEDUP

Online Shopping (E-Commerce) System

A Project Presentation | MERN Stack Web Application

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Live Project: https://stockedup.vercel.app

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Project Overview

Stockedup is a full-stack e-commerce website. Customers can browse products, add them to the cart and place orders, while an admin dashboard is used to manage the store. Built with the MERN stack.

- Two parts of the application - customer storefront and admin dashboard
- Product catalogue with 168 products in 6 categories, with search and filters
- Login/signup using JWT tokens; passwords stored in encrypted (hashed) form
- Cart, coupon-based discounts and order placement with e-mail confirmation
- Responsive design, works on mobile as well as desktop
- Hosted on Vercel, database on MongoDB Atlas (cloud)

Problem Statement

Why we built this project -

1) Small/local shops have a limited reach - they can only sell to nearby customers

2) Using existing marketplaces means paying high commission and platform fees

3) Customers waste time on slow search and unclear product information

4) Order and inventory handling is done manually, which is slow and error prone

5) Online stores are common targets of hacking - passwords and data need protection

6) There was no simple and affordable e-commerce solution for smaller stores

Objectives

Main Objectives

- Build an easy-to-use online store for browsing, searching and buying products
- Provide separate access for customers and admins using secure login
- Allow cart management, coupon discounts and order placement with confirmation
- Give admins a dashboard to manage products, categories and orders

Technical Objectives

- REST API with clear separation of client and server code
- Secure password hashing and JWT-based sessions
- Responsive UI that works on all screen sizes
- Cloud deployment on Vercel with MongoDB Atlas

Final deliverables: working web application, source code package and project report

Technology Stack

We used the MERN stack - JavaScript is used on both frontend and backend.

Layer	Technology	Used for
Frontend	React 18 + React Router + Vite	User interface, pages, routing
Backend	Node.js + Express	REST API, logic, authentication
Database	MongoDB (Atlas) + Mongoose	Stores users, categories, products, orders
Security	JWT + bcrypt	Login tokens and password hashing
Hosting / Tools	Vercel, Git, NPM	Deployment, version control, packages

System Architecture

CLIENT React SPA - user interface, pages, API calls

JSON over HTTPS

APPLICATION SERVER Node.js + Express - REST API, auth, business logic

Mongoose queries

DATABASE MongoDB Atlas - users, categories, products, orders

Important API calls Products & search -> GET /api/products, /api/categories

Login & register -> POST /api/auth/login, /api/auth/register

Place order -> POST /api/orders (can apply coupon code)

Database Design

MongoDB is a document database. This project uses 4 collections -

collection: users

name, email
password (hashed)
role (user/admin)

collection: categories

name
description
image

collection: products

name, price
description, image
categoryId, stock

collection: orders

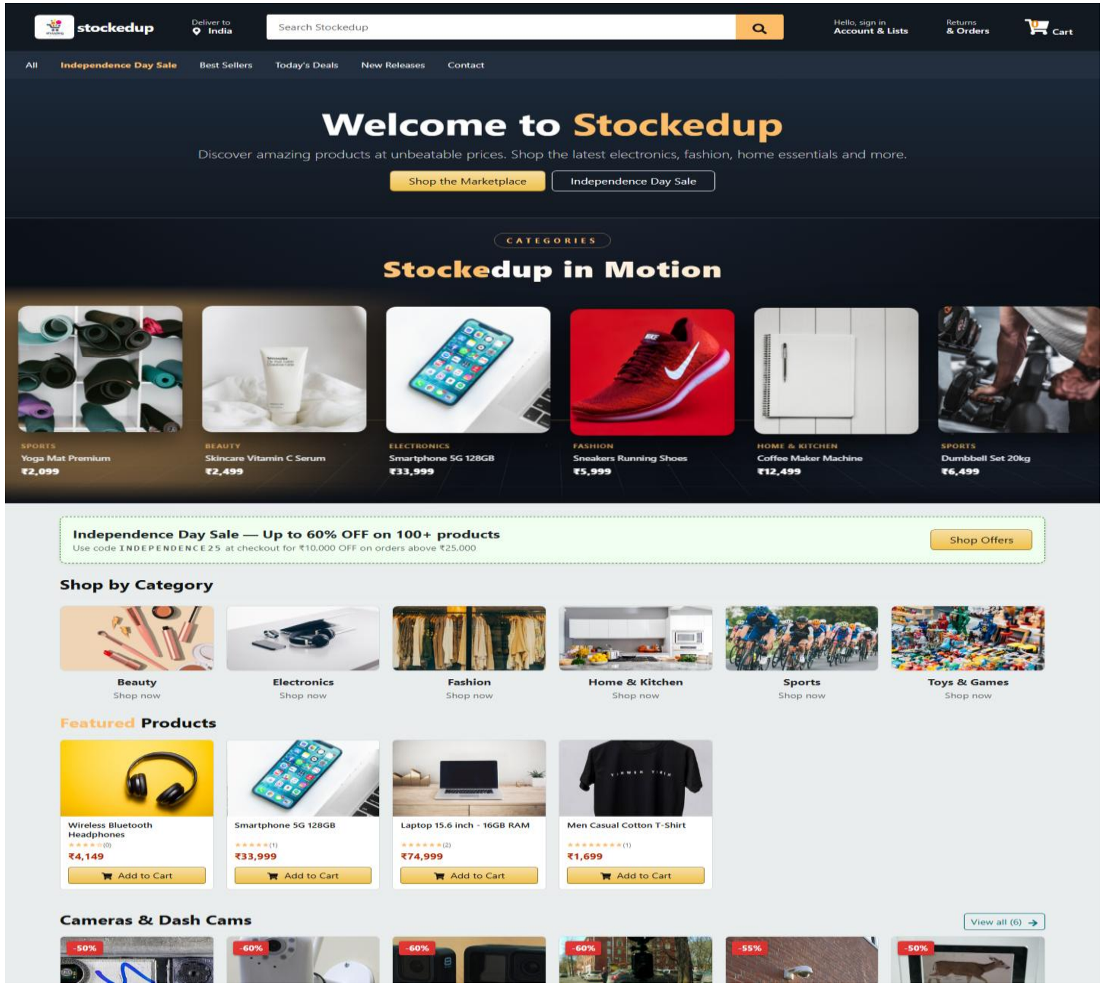
userId, items[]
subtotal, discount, total
couponCode, status

Relations: products.categoryId -> categories orders.userId -> users orders.items[] -> products

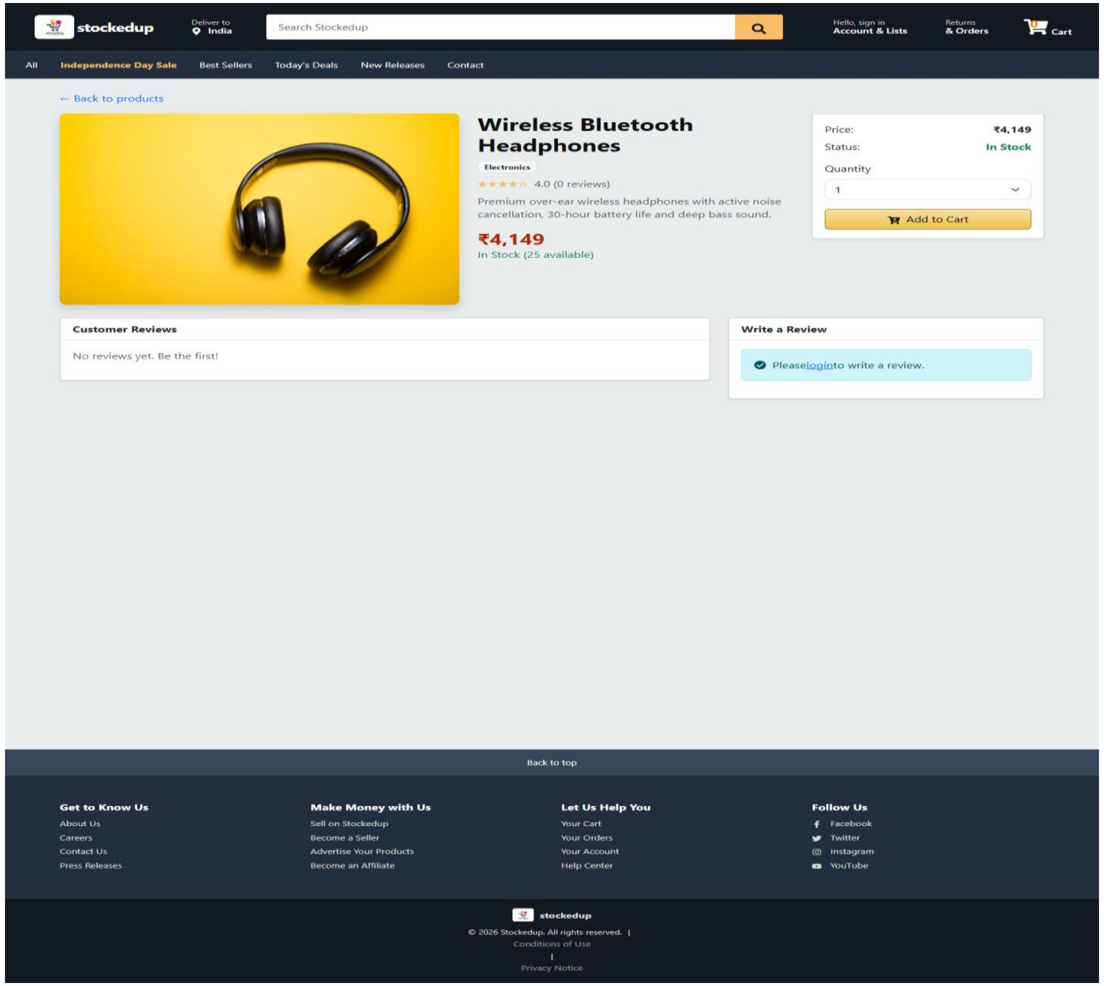
Main Modules

Module	Description
Authentication	Register / login for customers and admin, JWT based security
Product catalogue	Browse, search and filter 168 products in 6 categories
Cart & checkout	Add/remove items, apply coupon (INDEPENDENCE25), place order
Orders	Order placement with e-mail confirmation, admin can update status
Admin dashboard	Manage products and categories, view all orders
Other pages	Contact page, about, offers, 404 error page

Screenshots - 1

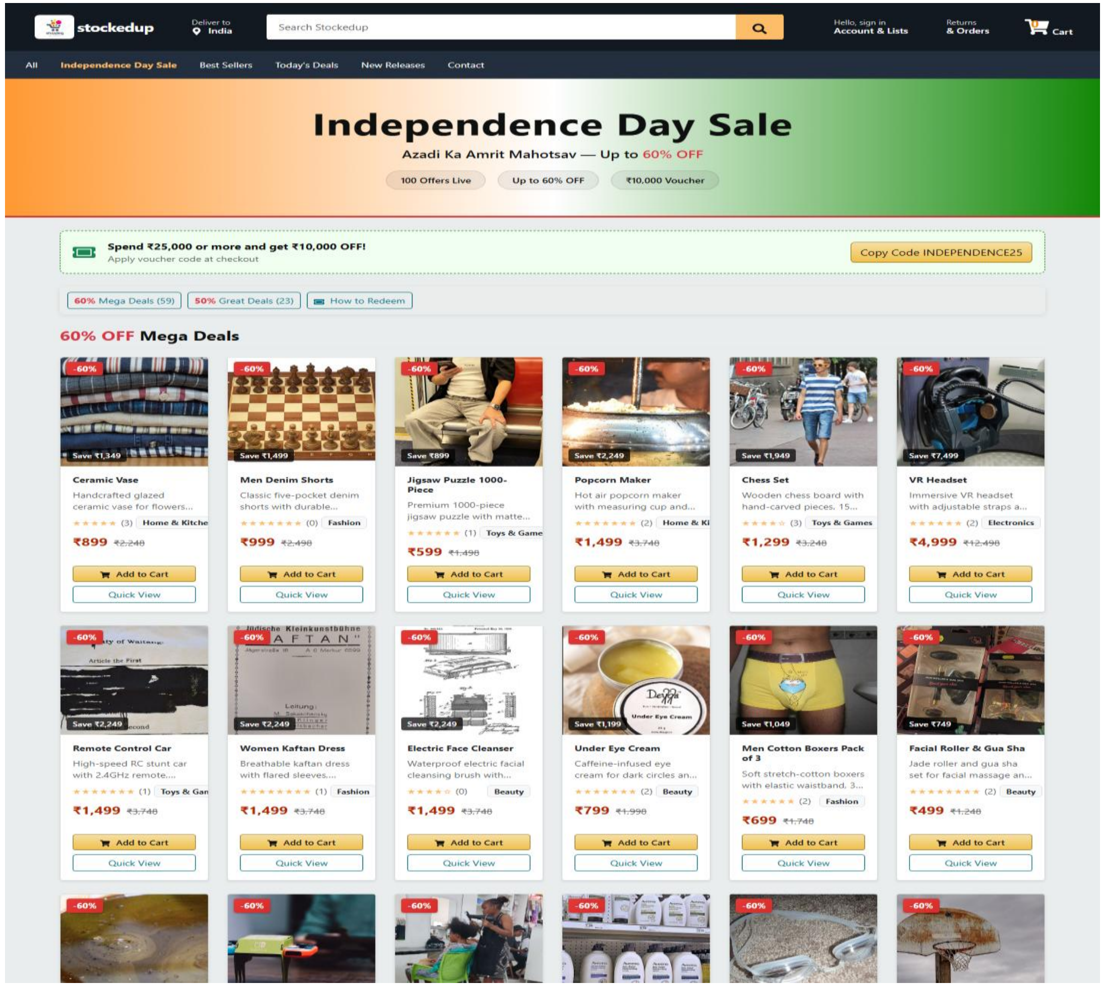


Home page - storefront

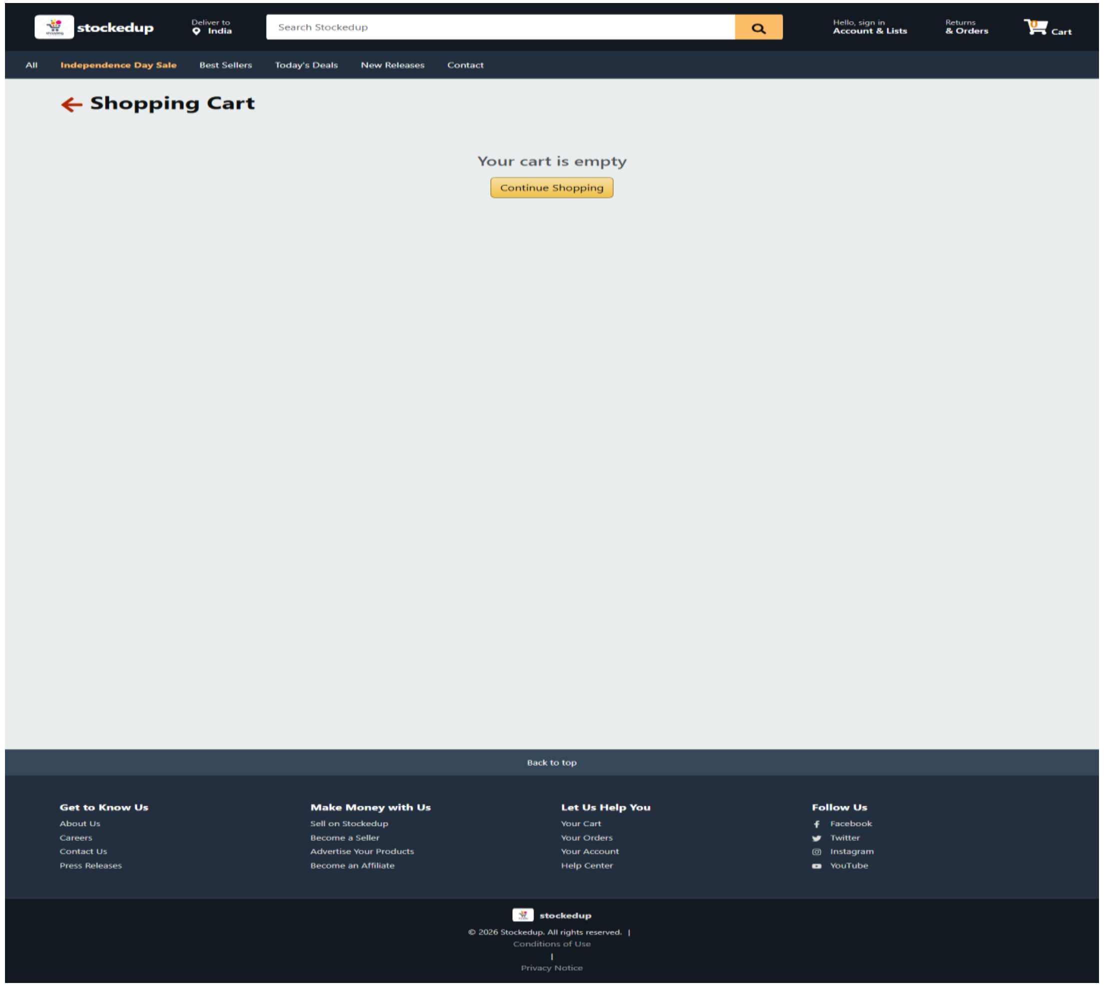


Product detail page

Screenshots - 2



Offers page with coupon INDEPENDENCE25



Cart page

Testing

- API testing - 16 test cases for auth, products, categories, cart, coupons and orders - all passed
- Functional testing - 10 user journeys tested on storefront and admin dashboard - all passed
- Responsive testing - checked the layout on mobile, tablet and desktop sizes
- Security testing - wrong passwords, fake tokens and unauthorised admin access are rejected
- Fixed bugs - carousel fallback when images fail, zoom issue on iPhone (small input font), slow page loading due to large images

Overall: all test cases passed before final deployment.

Conclusion & Future Scope

Conclusion

The project was completed successfully. Stockedup works as a full e-commerce website - customers can shop and place orders, and admins can manage the store. The application is live on Vercel and was tested thoroughly before submission.

Future Scope

- Online payment gateway (UPI / cards) for real transactions
- Order tracking, wishlist, reviews and ratings for products
- Better inventory management, stock alerts and sales reports
- Cloud storage for product images
- Installable mobile app support (PWA)

THANK YOU

Any questions?

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